

CS471 – Lesson 23 In-Class Worksheet

Temporal Reasoning: Hidden Markov Models

Part 1: Guided In-Class Worksheet (Instructor-Led)

Objective: Differentiate between observable and hidden temporal systems, execute the two-step forward filtering cycle, and interpret sequence inferencing tasks.

1. Observable vs. Hidden State Architectures

- a. Explain the fundamental difference between a standard Markov Chain and a Hidden Markov Model (HMM) in terms of state visibility.
- b. State the two conditional independence assumptions enforced by the graph topology of an HMM:
 - State Transition Assumption: _____
 - Sensor Emission Assumption: _____

2. Two-Step Forward Filtering Mechanics

An adversary surface radar transitions between two hidden states: $X_t \in \{\text{Search (S)}, \text{Track (T)}\}$. An Electronic Support Measures (ESM) receiver detects pulse emissions $E_t \in \{\text{Low (L)}, \text{High (H)}\}$.

- Prior Belief: $B(X_0) = \langle P(S) = 0.80, P(T) = 0.20 \rangle$
- Transitions: $P(S | S) = 0.70, P(T | S) = 0.30; \quad P(S | T) = 0.10, P(T | T) = 0.90$
- Emissions: $P(L | S) = 0.80, P(H | S) = 0.20; \quad P(L | T) = 0.10, P(H | T) = 0.90$

At time $t = 1$, the ESM sensor intercepts a High PRF pulse: $E_1 = H$.

- a. **Prediction Step:** Compute the time-projected belief distribution $\bar{B}(X_1) = \langle \bar{B}(S), \bar{B}(T) \rangle$.

- b. **Update Step:** Incorporate evidence $E_1 = H$. Calculate the unnormalized joint probabilities, the marginal evidence $P(E_1 = H)$, and the normalized belief state $B(X_1)$.

Part 2: Student Independent Practice

Objective: Apply HMM filtering, identify inference query types, and compute Viterbi trellis paths.

A. Submarine Acoustic Tracking

A naval sonobuoy field tracks a diesel-electric submarine operating in two hidden propulsion modes: $X_t \in \{\text{Battery (B)}, \text{Diesel (D)}\}$.

- Initial Prior at $t = 0$: $B(X_0) = \langle P(B) = 0.60, P(D) = 0.40 \rangle$
- State Transition Matrix:

	$X_t = \text{Battery}$	$X_t = \text{Diesel}$
$X_{t-1} = \text{Battery}$	0.80	0.20
$X_{t-1} = \text{Diesel}$	0.30	0.70

- Sonobuoy Hydrophone Sensor Model:

	$E_t = \text{Faint (F)}$	$E_t = \text{Loud (L)}$
$X_t = \text{Battery}$	0.90	0.10
$X_t = \text{Diesel}$	0.20	0.80

At time $t = 1$, the hydrophone records a Loud acoustic profile ($E_1 = L$).

1. Compute the predicted belief state $\bar{B}(X_1)$ prior to observing the acoustic measurement.
2. Compute the unnormalized likelihood-weighted probabilities given $E_1 = L$.
3. Compute the final normalized filtered belief $B(X_1) = P(X_1 \mid E_1 = L)$.

B. Inferential Task Classification

For each operational question below, identify whether the query corresponds to **Filtering**, **Smoothing**, or **Decoding**:

- a. An airborne battle management system determines whether an enemy SAM site is currently locking onto a strike aircraft at the present moment t : _____

- b. Intelligence analysts review flight data after a mission to determine the exact trajectory of an adversary drone across its entire 4-hour flight: _____
- c. Forensic cyber investigators calculate the probability that an unauthorized admin escalation took place at 0300 hours given all logs collected through 2400 hours: _____